

Technical Skills

Languages	C#, Kotlin, Java, TypeScript, JavaScript
Libraries / Frameworks	Angular, React, Node, .NET, Web API, Entity Framework, Dapper
Databases	MongoDB, SQL Server, PostgreSQL, DynamoDB, CosmosDB
Systems / Platforms	GCP, AWS, Azure, Kubernetes, Docker, Helm

Work Experience

Oct '23 - Present	Software Engineer - TAINA Tech Ltd. <i>TAINA Validator</i> - Led the containerisation of a .NET 6 Web API and React project with Docker, substantially improving development workflows and deployment consistency. Expertly implemented Kubernetes in Amazon EKS for application orchestration, utilising Helm charts for streamlined deployment and management of services. This strategic integration fostered a scalable, resilient hosting solution on AWS, optimising operational efficiency and minimising infrastructure overhead - Transformed database migration processes by transitioning from manual scripts to implementing FluentMigrations, automating migrations to execute upon project startup. This innovation significantly streamlined database updates, enhancing consistency and reducing manual workload - Initiated and executed a comprehensive refactoring of the API project, focusing on the efficient management of service lifetimes. This pivotal effort significantly elevated the application's performance and streamlined maintenance processes - Identified a critical SQL Server connection leak in the first week at the company, causing severe performance slowdowns for customers using SQL Server - Prioritised the transition from end-to-end UI testing to integration testing using programmatic interfaces, significantly speeding up test and deployment processes while enhancing test robustness - Mentored junior team members through fortnightly sessions, focusing on clean code practices and encouraging them to extend their capabilities beyond their comfort zones. These mentorship activities aimed to foster professional growth, enhance code quality, and build a culture of continuous learning and improvement within the team
Oct '22 - Sep '23	Software Engineer - Dojo Ltd. <i>Communications & Identity</i> - Worked alongside a compact and nimble team to create and sustain a shared service platform, streamlining the process of crafting services for in-house teams. These services were implemented in .NET 7, Kubernetes, and Docker, utilising a CQRS architectural approach and employing MongoDB - Successfully implemented an event-driven architecture on Google Cloud Platform (GCP) using PubSub, enhancing the platform's scalability and responsiveness, using a Domain-Driven Design - Participated in the upgrade initiative for all services that leverage our service, enabling them to embrace the latest version of our library and PubSub event model, and successfully transitioning all services to .NET 7. This expedited the retirement of the legacy system, resulting in a rapid and efficient transition - Part of an initiative to redesign the core internal service, consolidating it into a centralised instance that served as the primary point of communication for all other services, eliminating the need for ephemeral environments - Enhanced the internal tooling websites by incorporating additional features for various teams, utilising both Angular and React frameworks - Engaged in a novel project focused on implementing an in-house Identity service alongside the existing legacy service, which is in the process of being phased out. Played a role in shaping architectural choices, integrating Auth0, and managing project schedules - Conducted database migration operations to modernise legacy databases by transferring historical storage data, converting it into the updated data model, and seamlessly integrating it into the new database infrastructure, thereby ensuring the synchronisation and advancement of the new database system
Oct '21 - Aug '22	Software Engineer - HealthHero Ltd. <i>MedSol PWA / SAPI</i> - Migrated old monolithic front-end repository into a monorepo solution, and rewrote the various client implementations into individual PWAs written in React - Carried out a project to introduce an adequate test framework into a .NET API named SAPI which had little-to-no testing in place, resulting in a Unit Test coverage of above 80% and an Integration Test coverage of above 70% - Contributed to the development of a specialised database migration tool tailored for Azure CosmosDB, enabling the anonymisation of personally identifiable information (PII) data to ensure compliance with GDPR regulations while facilitating data analysis capabilities for cross-functional teams

Projects

Automated MMORPG Scripts

As a passionate gamer, my curiosity led me to explore the creation and execution of automated scripts in online games. During my exploration, I came across an open platform where I had the opportunity to engage with a deobfuscated version of the game Runescape, and interact with an open-source API developed for it. You can find the plugins related to this exploration on my GitHub page here: [harvey-plugins](#)

AWS Stock Checker

Created a straightforward, functional AWS script designed to monitor the availability of a specific item on a designated website and automatically notify the user via email when the item becomes available. Developed this project in response to a friend's request for a custom notifier solution when faced with the absence of built-in notification options

Artificial Neural Network for Intrusion Detection

Final year project to develop a Neural Network for aid in intrusion detection, using the NSL-KDD dataset. Trained the Network using a back-propagation method to detect intrusions by analysing packet attributes. Programmed and trained the network in MATLAB and resulted in an 87% detection rate. Paper available at [harveysingh.co.uk/NeuralNetwork.pdf](#), and files available on request

Android Application Development

A group project to develop an Android application. Elected as team leader by my fellow members; gaining leadership experience in a project environment. Produced a set of deadlines within the group to ensure prioritised tasks were completed within set targets. Carried out research using journals, articles, the Internet and IEEE to ensure that the application met the "golden-rules" of user-friendliness. Developed the application using Java; using MySQL to develop the back-end database

Java Finch Robot Application

A group project to control a Finch Robot, with heavy use of the Strategy Design Pattern to complete this task. Designed and implemented a test-plan; ensuring full functionality was met and checking for any errors before the project was. Conducted a demonstration of the robot followed by a code review to analyse my methodology, which was met with a lot of praise for my effective use of the command pattern. Source code available at <https://github.com/harveysingh1/Finch-Robot-Project>

HarveySingh.co.uk, my personal website

I am in the process of (re)building [harveysingh.co.uk](#), a website which will contain information about myself and my history. Currently the website is written in pure HTML, CSS, JavaScript, with the responsive layout being design using Bootstrap. I am going to be remaking the whole website using Angular, whilst also creating a blog within which I will hope to share any relevant industry knowledge I have (and will) gain over the coming years of my career

Education

2012-2017	Bachelor of Science in Computer Science Brunel University , London	Award: Second Upper Class (2:1)
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About Me

- One of the biggest hobbies I have is modifying and repairing cars. I recently have purchased a classic E30 BMW with hopes to do an engine conversion followed by fitting a turbocharger and several other upgraded components
- Programming and learning new technologies has now become a big part of my life, and I try to keep myself up-to-date with the latest in tech by attending conferences, joining meetup groups, and going to training courses
- I consider myself quite a social person and enjoy group outings with my team; I frequently go to escape rooms and other group activities which I think is beneficial for a team to mesh together
- I am an avid gamer and find myself playing the latest Triple-A titles and even older games such as Classic World of Warcraft and Runescape. I have found a new interest in automating the "grind" needed in some of these games, and have attempted to create scripts in order to automate these processes
- I like to keep fit and help others reach their fitness goals. So far I have lost approximately 36kg and have no plans to stop yet! I have taught many friends and family about the best (and healthiest) ways to reach their goals, whether they are trying to gain mass or lose weight, with successful results
- My work at ASOS and Veritas introduced me to Artificial Intelligence (specifically Natural Language Processing) has piqued my interest in the advancing region of Machine Learning and AI in general. I hope to work with this area of technology in my future career

References available upon request